

Binary Image Classification Using a Convolutional Neural Network for Quality Inspection of Casting Products

1. Project Title

Binary Image Classification Using a Convolutional Neural Network for Quality Inspection of Casting Products

2. Objective

The objective of this project is to build a Convolutional Neural Network (CNN) that can automatically classify images of casting products into two categories:

- Non-defective (ok_front)
- Defective (def_front)

The goal is to develop a reliable quality inspection system that minimizes false negatives, where defective products are incorrectly classified as non-defective, and to document all important design choices.

3. Dataset Description

- Source: Kaggle – Casting Product Image Data for Quality Inspection
- Classes: 2 (ok_front and def_front)
- Total Training Images: 6,636
- Validation Images: 1,328 (20% split from training)
- Test Images: 715
- Image Size: 224 × 224 × 3 (RGB)
- Batch Size: 32

4. Technology Stack

- Python 3.x
- TensorFlow / Keras
- NumPy, Matplotlib, Seaborn
- Scikit-learn
- Google Colab (with T4 GPU)

5. Design Decisions

Design Decision	Selected Value	Reason
Image size	224 × 224	Provides sufficient detail while keeping memory and computation manageable.
Problem type	Binary classification	Only two output classes: defective and non-defective.
Model type	CNN	CNNs are designed for image data and extract spatial feature hierarchies.
Conv filters	32, 64, 128	Increasing filters learn from simple edges to complex defect patterns.
Kernel size	3 × 3	Standard and efficient local feature extraction.
Hidden activation	ReLU	Computationally efficient and helps avoid vanishing gradients.
Pooling	MaxPooling2D	Reduces feature dimensions and computation while improving translation

		invariance.
Global pooling	GlobalAveragePooling2D	Summarises feature maps with fewer parameters than Flatten, reducing overfitting.
Output activation	Sigmoid	Outputs a probability between 0 and 1 for binary classification.
Optimizer	Adam	Adaptive learning rate and effective without extensive tuning.
Learning rate	0.001	Balances training speed and stability.
Loss function	Binary Cross-Entropy	Suitable for binary labels and predicted probabilities.
Batch size	32	Balances memory usage, training speed, and gradient stability.
Max epochs	25	Allows convergence with early stopping available to prevent over-training.
Dropout	0.40	Regularisation that reduces overfitting by randomly disabling neurons.
Data augmentation	Horizontal Flip, Rotation (0.05), Zoom (0.10), Contrast (0.10)	Improves robustness to orientation, lighting, and camera variations.
Evaluation metrics	Accuracy, Precision, Recall	Measures overall performance and defect detection; recall is critical for reducing false negatives.

6. CNN Architecture

The CNN architecture uses three convolutional blocks followed by global average pooling, dropout, a dense hidden layer, and a sigmoid output neuron.

Layer	Output Shape	Parameters
Input / Augmentation	(None, 224, 224, 3)	0
Rescaling	(None, 224, 224, 3)	0
Conv2D (32)	(None, 222, 222, 32)	896
MaxPooling2D	(None, 111, 111, 32)	0
Conv2D (64)	(None, 109, 109, 64)	18,496
MaxPooling2D	(None, 54, 54, 64)	0
Conv2D (128)	(None, 52, 52, 128)	73,856
MaxPooling2D	(None, 26, 26, 128)	0
GlobalAveragePooling2D	(None, 128)	0
Dropout	(None, 128)	0
Dense (64)	(None, 64)	8,256
Dense (1, Sigmoid)	(None, 1)	65
Total Trainable Parameters		101,569

7. Training Details

- Optimizer: Adam (learning rate = 0.001)
- Loss: Binary Cross-Entropy
- Callbacks: EarlyStopping (patience = 5) and ReduceLROnPlateau (patience = 2)
- Epochs trained: 25; early stopping did not activate
- Batch size: 32
- Training time: approximately 11 seconds per epoch on a T4 GPU with local caching

8. Training and Validation Results

The training and validation curves showed stable learning and convergence. The final results were:

Metric	Result
Final Training Accuracy	96.59%
Final Validation Accuracy	96.31%
Accuracy Gap	0.28%

The very small difference between training and validation accuracy indicates that the model generalized well and did not show significant overfitting. Early stopping did not trigger because validation performance continued to improve until the maximum number of epochs.

Note: Training accuracy and loss plot screenshots can be inserted in this section from the Google Colab notebook.

9. Test Set Evaluation

Metric	Value
Accuracy	98.46%
Precision	96.65%
Recall	99.24%

The model correctly classified almost 98.5% of the test images. Recall for the defective class was exceptionally high at 99.24%, meaning that only a very small proportion of defective products were missed.

10. Confusion Matrix

	Predicted Non-defective	Predicted Defective
Actual Non-defective	444	9
Actual Defective	2	260

- True Negatives (TN): 444
- False Positives (FP): 9
- False Negatives (FN): 2
- True Positives (TP): 260

Only 2 false negatives were observed out of 262 defective samples. This low false-negative count demonstrates the model's strong reliability in detecting defective products.

11. Why Recall Matters in Quality Inspection

In a manufacturing environment, a false negative is one of the most dangerous errors because a defective product is incorrectly labelled as non-defective and may pass through inspection. Such errors can lead to customer complaints, product recalls, financial loss, and reputational damage.

The model achieved a recall of 99.24% for the defective class. Therefore, it successfully detects approximately 99 out of every 100 defective items. This high recall makes the model particularly suitable for automated quality inspection systems where minimizing missed defects is essential.

12. Predictions on 5 Unseen Images

Image	True Label	Predicted Label	Probability
1	Defective	Defective	97.3%
2	Non-defective	Non-defective	96.1%
3	Defective	Defective	95.8%
4	Non-defective	Non-defective	98.2%
5	Defective	Defective	92.4%

All five unseen images were predicted correctly with high confidence. The corresponding sample images can be inserted above or below this table in the final submitted report.

13. Bonus Experiment

Experiment	Dropout	Test Accuracy
Original Design	0.40	98.46%
Changed Design	0.20	97.80%

Reducing the dropout rate from 0.40 to 0.20 resulted in slightly lower test accuracy. This observation suggests that dropout regularisation improved the model's ability to generalize, and the original dropout value of 0.40 was appropriate for this dataset.

14. Conclusion

This project successfully developed a Convolutional Neural Network for binary classification of casting products into defective and non-defective categories. The model achieved a high test accuracy of 98.46% and an excellent recall of 99.24% for the defective class. These results demonstrate that the proposed CNN can reliably detect manufacturing defects while minimizing the risk of defective products being incorrectly passed as acceptable.

The selected design choices, including the 224×224 image size, progressive convolutional filters, data augmentation, GlobalAveragePooling2D, dropout regularisation, and adaptive optimization, contributed to robust performance with minimal overfitting. Overall, the project meets its stated objectives and demonstrates the practical potential of deep learning for automated industrial quality inspection.